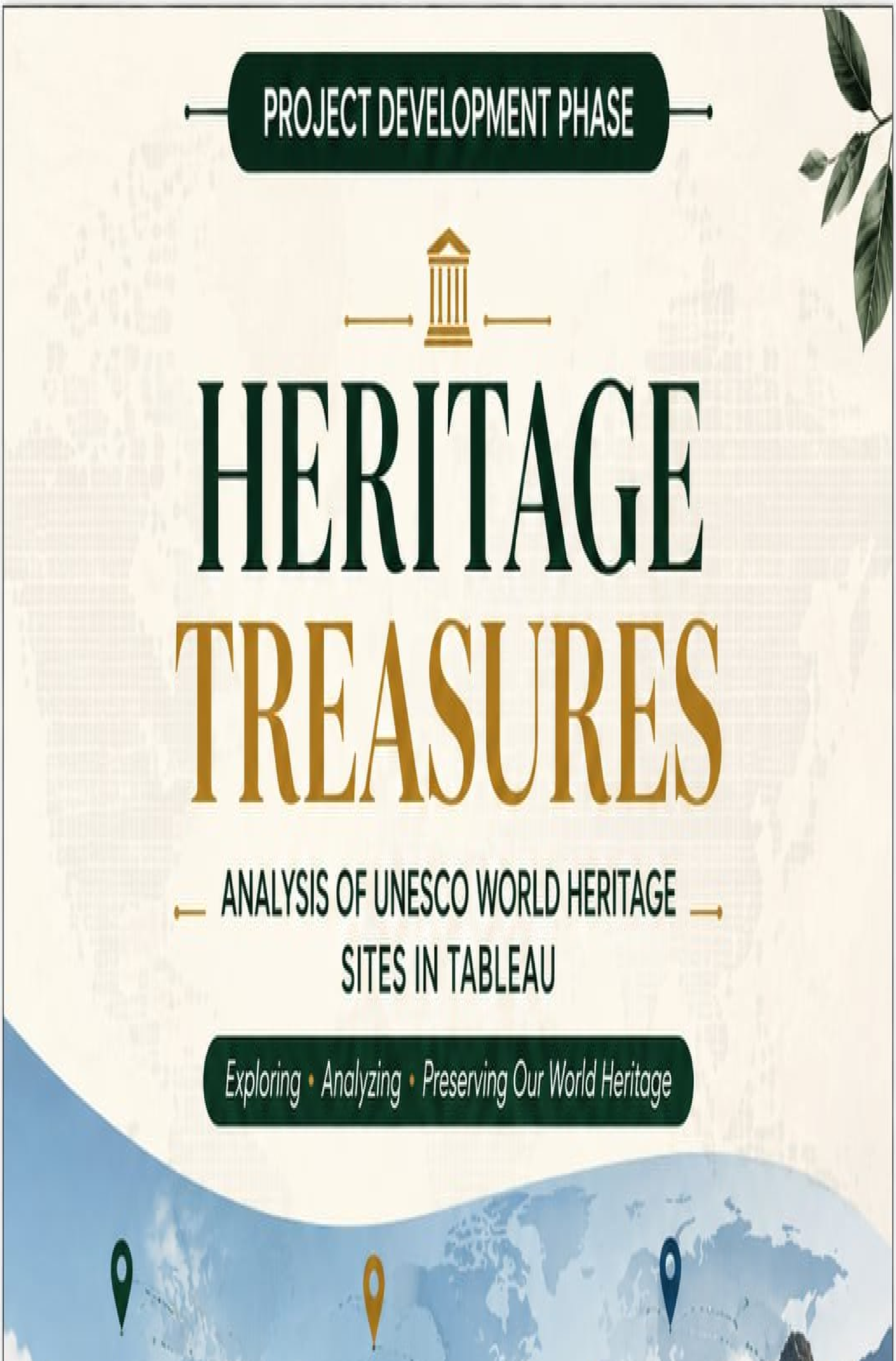
PROJECT DEVELOPMENT PHASE
HERITAGE
TREASURES
ANALYSIS OF UNESCO WORLD HERITAGE
SITES IN TABLEAU
Exploring • Analyzing • Preserving Our World Heritage

Data Processing Steps :

Step 1: Import the Dataset

The first step in the project development process was to import the UNESCO World Heritage Sites dataset into Tableau. The dataset was available in CSV/Excel format and contained comprehensive information about heritage sites across different countries and regions.

During the import process, all dataset fields were verified to ensure that they were correctly loaded into Tableau. Important attributes such as Site Name, Country, Region, Category, Year Inscribed, Danger Status, Latitude, Longitude, and Area (Hectares) were checked for completeness.

The imported dataset served as the foundation for the dashboard development process. Successful data import ensured that further preprocessing and visualization activities could be performed without errors.

Activities Performed

- Imported UNESCO World Heritage Dataset.
- Verified all dataset columns.
- Checked data loading status.
- Confirmed successful Tableau connection.

Expected Output

A complete and accessible dataset ready for preprocessing.

Step 2: Check Missing Values

After importing the dataset, the next step was to identify missing or null values that could affect the accuracy of the analysis.

Each column was carefully examined to determine whether important information was missing. Special attention was given to fields such as Country, Region, Category, Year Inscribed, Latitude, Longitude, and Danger Status because these fields are essential for visualization.

Wherever missing values were found, appropriate preprocessing techniques were applied. Records with unnecessary missing values were cleaned or standardized to maintain data quality.

This process improved the reliability and consistency of the dataset.

Activities Performed

- Checked all columns for null values.
- Identified incomplete records.
- Corrected missing information wherever possible.
- Ensured no important analytical field remained empty.

Expected Output

A clean dataset with minimal missing values.

Step 3: Remove Duplicate Records

Duplicate records can lead to incorrect statistical results and misleading visualizations. Therefore, duplicate heritage site entries were carefully identified and removed.

The dataset was checked using unique identifiers such as Site Name and Country combination to ensure that every UNESCO Heritage Site appeared only once.

After duplicate removal, the total number of records was verified again to confirm that the dataset accurately represented all heritage sites.

Activities Performed

- Checked duplicate site names.
- Compared duplicate country records.
- Removed repeated entries.
- Verified final record count.

Expected Output

A duplicate-free dataset.

Step 4: Validate Data Types

Correct data types are essential for accurate Tableau analysis. Therefore, every field was examined and assigned the appropriate data type.

Field	Data Type
Site Name	String
Country	Geographic
Region	String
Category	String
Year Inscribed	Number
Latitude	Number
Longitude	Number
Area (Hectares)	Decimal
Danger Status	String

Correct data types ensured that maps, filters, charts, and calculations worked properly.

Activities Performed

- Verified all data types.
- Converted incorrect data types.
- Validated geographic fields.
- Tested numeric fields.

Step 5: Verify Data Consistency

Data consistency ensures that similar values follow the same naming convention throughout the dataset.

For example, heritage categories were standardized as:

- Cultural
- Natural
- Mixed

Similarly, country names and region names were checked for spelling inconsistencies, unnecessary spaces, or duplicate naming patterns.

This step improved the overall quality of visualizations and prevented duplicate categories from appearing in charts.

Activities Performed

- Standardized country names.
- Standardized region names.
- Corrected category values.
- Removed formatting inconsistencies.

Step 6: Prepare Calculated Fields

Several calculated fields were created in Tableau to simplify dashboard development and improve analytical capabilities.

Calculated Fields Used

Calculated Field	Purpose
Total Heritage Sites	Count all heritage sites
Cultural Sites	Count cultural sites
Natural Sites	Count natural sites
Mixed Sites	Count mixed sites
Sites by Country	Country-wise comparison
Sites by Region	Region-wise comparison

Year-wise Growth	Trend analysis

These calculated fields were later used to create KPI Cards, charts, filters, and interactive dashboards.

Step 7: Final Clean Dataset

After completing all preprocessing activities, the dataset was verified one final time.

Quality checks confirmed that:

- No duplicate records remained.
- Missing values were handled.
- Data types were correct.
- Categories were standardized.
- Geographic fields were properly configured.

The final cleaned dataset was then connected to Tableau worksheets to build interactive dashboards and stories.

Final Output

A high-quality, clean, and analysis-ready UNESCO World Heritage dataset suitable for dashboard development.

Business Questions & Visualization Analysis

Business Question 1

What is the total number of UNESCO World Heritage Sites across the world?

Objective

The objective of this analysis is to determine the total number of UNESCO World Heritage Sites available worldwide. This KPI provides a quick overview of the overall size of the UNESCO heritage database and serves as the primary performance indicator of the dashboard.

Visualization Used

KPI Card

Analysis

A KPI Card displays the total count of UNESCO World Heritage Sites in a single, easy-to-read metric. This allows users to immediately understand the scale of heritage sites without navigating through multiple reports.

Expected Outcome

Users can quickly identify the total number of heritage sites included in the dataset.

Business Question 2

Which countries have the highest number of UNESCO World Heritage Sites?

Objective

The purpose of this analysis is to compare different countries based on the number of UNESCO World Heritage Sites they possess.

Visualization Used

Horizontal Bar Chart

Analysis

The countries are arranged in descending order based on the total number of heritage sites. This helps identify countries with the richest cultural and natural heritage.

The visualization also allows easy comparison among countries.

Expected Outcome

Users can identify the top countries contributing the highest number of UNESCO heritage sites.

Business Question 3

How are UNESCO Heritage Sites distributed across different regions of the world?

Objective

The objective is to analyze the geographical distribution of UNESCO Heritage Sites across various regions.

Visualization Used

Filled Map (World Map)

Analysis

The filled map highlights countries and regions according to the number of heritage sites. Different shades indicate varying concentrations of heritage locations.

This visualization enables users to easily understand the global distribution of UNESCO sites.

Expected Outcome

Users can identify which regions have the highest and lowest concentration of heritage sites.

Business Question 4

What is the distribution of Cultural, Natural, and Mixed Heritage Sites?

Objective

The objective is to understand how UNESCO heritage sites are categorized.

Visualization Used

Pie Chart

Analysis

The pie chart represents the percentage share of Cultural, Natural, and Mixed Heritage Sites.

This visualization provides an overview of category distribution and highlights which heritage type dominates globally.

Expected Outcome

Users can easily compare different heritage categories.

Business Question 5

How has the number of UNESCO World Heritage Sites changed over the years?

Objective

To study historical growth and trends in UNESCO World Heritage Site inscriptions over time.

Visualization Used

Line Chart

Analysis

The line chart shows year-wise growth in the number of heritage sites. It helps identify periods of rapid increase or slower growth and provides insight into UNESCO's heritage recognition trends.

Expected Outcome

Users can understand the historical trend and growth pattern of UNESCO World Heritage Sites.

KPI Analysis

Introduction

The KPI section is displayed at the top of the Tableau dashboard and acts as a summary panel for decision-makers, researchers, tourists, students, UNESCO authorities, and government agencies.

KPI 1: Total UNESCO World Heritage Sites

Objective

The objective of this KPI is to display the total number of UNESCO World Heritage Sites available in the dataset. It provides an overall summary of the heritage database.

Formula

COUNT([Site Name])

Analysis

This KPI counts all UNESCO World Heritage Sites available in the dataset. It gives users an immediate understanding of the size of the heritage database and serves as the primary indicator of the dashboard. This metric is useful for evaluating the overall coverage of the analysis.

KPI 2: Total Countries Covered

Objective

To determine the total number of countries that contain UNESCO World Heritage Sites.

Formula

COUNTD([Country])

Analysis

This KPI counts the number of unique countries represented in the dataset. It highlights the geographical coverage of UNESCO heritage sites and helps users understand the global reach of the analysis.

KPI 3: Cultural Heritage Sites

Objective

To display the total number of Cultural Heritage Sites.

Formula

IF [Category]="Cultural" THEN 1 END

Analysis

This KPI counts only Cultural Heritage Sites. It helps users understand the contribution of cultural heritage to the overall UNESCO database. Since cultural heritage forms the largest category, this KPI is useful for heritage researchers and historians.

KPI 4: Natural Heritage Sites

Objective

To display the total number of Natural Heritage Sites.

Formula

IF [Category]="Natural" THEN 1 END

Analysis

This KPI calculates the number of UNESCO sites categorized as Natural Heritage. It highlights biodiversity conservation and environmentally significant locations worldwide.

KPI 5: Mixed Heritage Sites

Objective

To display the total number of Mixed Heritage Sites.

Formula

IF [Category]="Mixed" THEN 1 END

Analysis

Mixed Heritage Sites possess both cultural and natural significance. This KPI identifies such sites and helps users understand their contribution to global heritage conservation.

KPI 6: Endangered Heritage Sites

Objective

To determine the total number of UNESCO World Heritage Sites currently classified as endangered.

Formula

COUNT(IF [Danger Status]="Yes" THEN [Site Name] END)

Analysis

This KPI highlights heritage sites that require urgent conservation efforts. It enables researchers, UNESCO authorities, and governments to focus on vulnerable heritage sites and monitor conservation priorities.

Dashboard Summary

Heritage Treasures: Analysis of UNESCO World Heritage Sites in Tableau

Overview

The Global Heritage Sites Analysis Dashboard is an interactive Tableau dashboard designed to transform UNESCO World Heritage data into meaningful visual insights. The dashboard provides a centralized platform where users can explore, compare, and analyze heritage sites across different countries, regions, and categories. It enables researchers, students, tourists, UNESCO authorities, and government agencies to gain valuable insights through dynamic visualizations instead of lengthy reports.

The dashboard combines multiple visual elements such as KPI Cards, World Maps, Bar Charts, Pie Charts, Line Charts, Treemaps, and Interactive Filters to provide a comprehensive analysis of UNESCO World Heritage Sites. Users can easily filter the data based on countries, regions, heritage categories, and danger status, making the dashboard highly interactive and user-friendly.

Dashboard Components

1. KPI Cards

The dashboard begins with KPI Cards that provide a quick overview of the dataset. These cards display important statistics such as the total number of UNESCO World Heritage Sites, total countries represented, category-wise site counts, and endangered heritage sites. KPIs allow users to understand the overall dataset at a glance.

2. World Map Visualization

The interactive World Map displays the geographical distribution of UNESCO World Heritage Sites across different countries. Users can quickly identify countries with a high concentration of heritage sites and explore global heritage distribution through map-based visualization.

3. Country-wise Analysis

A horizontal bar chart compares countries based on the number of UNESCO World Heritage Sites. This visualization helps identify the top-performing countries and supports comparative analysis.

4. Category-wise Analysis

A pie chart presents the distribution of Cultural, Natural, and Mixed Heritage Sites. This visualization helps users understand the contribution of each heritage category to the overall UNESCO heritage database.

5. Region-wise Analysis

A treemap or bar chart illustrates the number of heritage sites across different geographical regions. It enables users to compare regional contributions and identify areas with significant cultural and natural heritage.

6. Year-wise Trend Analysis

A line chart shows the growth of UNESCO World Heritage Site inscriptions over the years. This analysis helps users understand historical trends and observe how the number of recognized heritage sites has changed over time.

7. Endangered Heritage Sites Analysis

A dedicated visualization highlights UNESCO World Heritage Sites currently classified as endangered. This feature assists researchers and policymakers in identifying vulnerable heritage locations requiring conservation efforts.

8. Interactive Filters

The dashboard includes interactive filters for Country, Region, Heritage Category, and Danger Status. These filters allow users to customize the displayed information according to their requirements and perform detailed comparative analysis.

Story Summary

Heritage Treasures: Analysis of UNESCO World Heritage Sites in Tableau

The Story Dashboard presents the complete analysis of UNESCO World Heritage Sites in a structured and interactive sequence. Unlike individual dashboards that display isolated visualizations, the story connects multiple dashboards into a logical flow, enabling users to understand the complete heritage analysis step by step.

The Tableau Story was developed to guide users through different aspects of UNESCO World Heritage data, including global distribution, regional comparison, heritage categories, endangered sites, and historical trends. Each story point highlights a specific analytical perspective, making the dashboard easier to understand for researchers, students, tourists, UNESCO officials, and government agencies.

The story transforms complex data into an engaging narrative that supports informed decision-making and enhances the overall user experience.

Conclusion

Heritage Treasures: Analysis of UNESCO World Heritage Sites in Tableau

The Heritage Treasures: Analysis of UNESCO World Heritage Sites in Tableau project was successfully developed to provide an interactive and insightful platform for analyzing UNESCO World Heritage Sites across the world. The project transformed raw heritage data into meaningful visualizations that enable users to explore, compare, and understand heritage information in an efficient and user-friendly manner.

During the project development process, the UNESCO World Heritage dataset was carefully collected, cleaned, validated, and prepared for analysis. Various data preprocessing techniques such as handling missing values, removing duplicate records, validating data types, standardizing data, and creating calculated fields were performed to ensure the accuracy and reliability of the dataset. These